

COST-EFFECTIVE AND FLEXIBLE SOLUTIONS WITH PXI

Three different applications where PXI was an important part of the required test strategy and in some cases made the efforts easier

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Every year products and systems with more intelligence and operational convenience are available; usually, at a lower cost than previous models. The basic premise of market-driven economies is the need to create new products that people will purchase.

As technology progresses, test-technologies and their implementation strategies change as well. The testing requirements these days include factors such as increased simulation of sensors, fault-insertion, higher data rates and increased precision. All of these must be accomplished while simultaneously increasing test throughput and lowering costs. While this has always been a tough challenge for the industry, it seems to be a more daunting prospect to support today's technology.

Fortunately, the test and measurement (T&M) industry has been developing new solutions to address these needs. In many cases the solution has been PCI eXtension for instrumentation (PXI), which is a remarkably flexible platform. Over the last fourteen years, the 50-plus members of the PXI Systems Alliance (PXISA) have continued to innovate and create modules that can address the newest technologies.

Why PXI?

Many vendors, many more products, and lots of previous applications' successes are key reasons for PXI's success. But looking beyond, there are other reasons as well:

Choices in software. Many programming languages, including real-

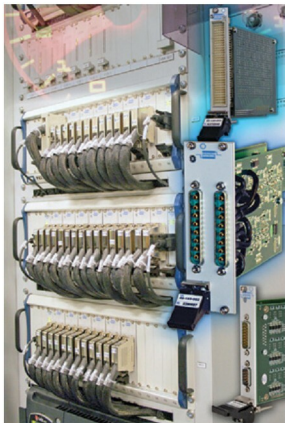


Fig. 1: Automotive hardware-in-the-loop simulation (HILS) test system

time operating systems as well as focused software for applications like hardware-in-the-loop simulation (HILS) make PXI implementation easier.

Connectivity. The best hardware is useless unless it can be connected to the unit-under-test (UUT). Fortunately, there are vendors who have created mass interconnects that work well with PXI.

Vendor creativity. The small size of 3U module (10×16 cm²) was seen as a detriment back in 1997 when the specification was released. It was felt that the small size, especially when

compared to versa module europa (VME) eXtensions for instrumentation (VXI), would limit the bandwidth and density that could be achieved. But the naysayers were wrong! Switching densities of 4000 to 8000 relays in a single matrix, 1000V isolation, and RF instrumentation of 6 GHz and greater are just a few examples of what is possible in PXI.

Ability to work with other test platforms. In many instances, hybrid test systems are the norm. This can be due to the availability of test equipment, test budgets forcing hardware reuse, and partial re-hosting/upgrades of existing test systems. PXI has shown to work well in these environments.

System integrators. This segment of the test industry has embraced the

PXI platform as a solution for many applications. Their ability to make PXI work is a big part of the success.

How PXI has been employed?

Let's look at applications where specialised simulation of a UUT's operating environment is necessary for test. Every test system is a simulator, as the system replicates the car, PC or the missile that the UUT will ultimately end up in. But new applications require new simulation techniques.